A Project report on

An IoT-Based Smart Doorbell System Using Camera and Ultrasonic Sensor for Real-Time Visitor Detection and Context-Aware Monitoring

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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Certificate

This is to certify that the project report entitled **An IoT-Based Smart Doorbell System Using Camera and Ultrasonic Sensor for Real-Time Visitor Detection and Context-Aware Monitoring** is the bonafide work carried out by **G. Thahaseen, K. Sowmya Latha, M. Simhadri Naik, P. Venkata Jagadeeswara Reddy,** bearing Roll Number **224G1A33B9, 224G1A33A3, 234G5A3309, 234G5A3312,** in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering (Artificial Intelligence & Machine Learning)** during the academic year 2025-2026.

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The results embodied in this project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

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ABSTRACT

The An IoT-Based Smart Doorbell System Using Camera and Ultrasonic Sensor for Real-Time Visitor Detection and Context-Aware Monitoring is designed to enhance home security, enable real- time monitoring, and provide controlled access to residential environments. Traditional doorbell systems lack intelligence, remote accessibility, and verification mechanisms, making them less secure and inefficient. This system leverages IoT technology along with embedded Python and Arduino-based control to detect visitors, capture images, and notify homeowners instantly through a web or mobile interface.

The system integrates components such as an ultrasonic sensor, USB camera, Arduino controller, and communication modules. When a visitor approaches, the ultrasonic sensor detects their presence and triggers the camera to capture an image, which is sent to the homeowner via Telegram for verification. Based on the received image, the user can grant or deny access remotely. Upon approval, the system activates the motor driver and voice module to unlock and open the door, which automatically closes after a preset duration, ensuring both convenience and security.

Additional safety features such as a GSM module, buzzer, and LED indicators provide alerts during unauthorized access or emergency situations. The system efficiently logs all activities, including visitor detection and access control, ensuring reliability and traceability. Overall, the proposed system offers a smart, cost-effective, and scalable solution suitable for modern homes, improving security while reducing manual effort through automation and remote monitoring.

Keywords: Home Security, Ultrasonic Detection, Remote Access, Telegram Automation, Smart Door Control, Embedded Python Control, Arduino-Based Security, IoT Home Automation.

PROJECT CONTRIBUTION TOWARDS SUSTAINABLE DEVELOPMENT GOALS

Project Title:

An IoT-Based Smart Doorbell System Using Camera and Ultrasonic Sensor for Real-Time Visitor Detection and Context-Aware Monitoring

SDG - Project Mapping:

S.No	Sustainable Development Goals	Mapped (Yes/No)
1	SDG 1: No Poverty	No
2	SDG 2: Zero Hunger	No
3	SDG 3: Good Health and Well-Being	Yes
4	SDG 4: Quality Education	Yes
5	SDG 5: Gender Equality	Yes
6	SDG 6: Clean Water and Sanitation	No
7	SDG 7: Affordable and Clean Energy	No
8	SDG 8: Decent Work and Economic Growth	Yes
9	SDG 9: Industry Innovation and Infrastructure	Yes
10	SDG 10: Reduced Inequalities	Yes
11	SDG 11: Sustainable Cities and Communities	Yes
12	SDG 12: Responsible Consumption and Production	No
13	SDG 13: Climate Action	No
14	SDG 14: Life Below Water	No
15	SDG 15: Life on Land	No
16	SDG 16: Peace, Justice and Strong Institutions	Yes
17	SDG 17: Partnerships for the Goals	No

Explanation:

- The project enhances home safety and personal security through real-time monitoring and visitor detection, supporting SDG 3 (Good Health and Well-Being).
- By integrating IoT technology and smart automation, the system promotes innovation in smart home infrastructure, aligning with SDG 9 (Industry, Innovation, and Infrastructure).
- The system enables safer living environments in residential areas, contributing to SDG 11 (Sustainable Cities and Communities).
- The project also promotes learning and awareness of IoT technologies among students and developers, contributing to SDG 4 (Quality Education).

- The project contributes to economic growth by encouraging the adoption of smart technologies and reducing reliance on traditional security systems, aligning with SDG 8 (Decent Work and Economic Growth).
- By ensuring secure monitoring and controlled access, the system strengthens safety and trust within communities, supporting SDG 16 (Peace, Justice, and Strong Institutions).
- The project also promotes learning and awareness of IoT technologies among students and developers, contributing to SDG 4 (Quality Education).

Expected Impact:

Parameter	**Impact**
Home Security	Enhanced safety through real-time visitor detection and remote access control
Convenience	Enables remote monitoring and door access via Telegram notifications
Automation	Reduced manual effort with automatic door operations
Safety Alerts	Immediate alerts using buzzer, GSM module, and LED indicators

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
HC-SR04	Ultrasonic Distance Sensor
MCU	Microcontroller Unit
LCD	Liquid Crystal Display
LED	Light Emitting Diode t
DC	Direct Current
AC	Alternating Current
UML	Unified Modeling Language
ER	Entity-Relationship
DFD	Data Flow Diagram
IOREF	Input/Output Reference Voltage
IoT	Internet of Things
USB	Universal Serial Bus
WI-FI	Wireless Fidelity
GSM	Global System for Mobile Communications
UNO	Arduino Uno Microcontroller Board
PWM	Pulse Width Modulation
SPL	Sound Pressure Level
TWL	Two-Wire Link
AREF	Analog Reference Voltage

CHAPTER 1

INTRODUCTION

The rapid growth of Internet of Things (IoT) technology has enabled the development of smart and secure home automation systems. Traditional doorbell systems lack real-time monitoring, remote access, and intelligent decision-making, which can lead to security issues and inconvenience. To address these limitations, an IoT-Based Smart Doorbell System is proposed that enhances home security through automation and remote control.

The system uses an ultrasonic sensor to detect visitors and a USB camera to capture images, which are sent to the homeowner via Telegram for verification . Based on user response, the door can be automatically opened using a motor driver and voice module. Additional features like GSM alerts, buzzer, and LED indicators improve safety and ensure quick response to unauthorized access.

1.1 Motivation

Traditional doorbell systems are commonly used in residential environments, but they have several limitations such as lack of real-time monitoring, absence of remote access, and inability to verify visitors before granting entry. These systems rely entirely on manual interaction, which can lead to security risks, delayed responses, and inconvenience, especially when homeowners are not physically present. Additionally, they do not provide any mechanism to handle unauthorized access or emergency situations effectively.

With the rapid advancement of Internet of Things (IoT) technology, there is a growing need for intelligent systems that offer enhanced security, automation, and remote accessibility. Modern homes require solutions that can provide real-time alerts, visitor verification, and controlled access from anywhere. IoT-based systems enable seamless communication between devices, ensuring efficient monitoring and quick response to potential threats.

Therefore, the motivation behind this project is to develop a smart doorbell system that integrates sensors, camera modules, and communication technologies to provide secure and automated home entry management. By enabling real-time visitor detection, remote verification, and automated door control, the system aims to improve home security, reduce manual effort, and provide a reliable and user-friendly solution for modern smart homes.

1.2 Problem Statement

Traditional doorbell systems rely on manual interaction and basic alert mechanisms, where the homeowner must be physically present to respond to visitors. These systems do not provide real-time monitoring, remote access, or visitor verification, which can lead to security vulnerabilities and inconvenience. The lack of intelligent features makes it difficult to identify unknown visitors and prevent unauthorized access.

Another major issue is the absence of automation and instant communication. Conventional systems do not offer image capture, remote notifications, or access control, resulting in delayed responses and limited control over home security. Additionally, there are no effective mechanisms to handle emergency situations or suspicious activities, increasing the risk of intrusion or unsafe conditions..

There is a need for a smart system that ensures real-time visitor detection, remote monitoring, and secure access control without requiring constant physical presence. IoT technology provides a solution by enabling connected devices to communicate and automate operations efficiently..

Hence, this project proposes an IoT-Based Smart Doorbell System that enhances home security, improves response time, and ensures reliable and automated entry management by integrating sensors, camera modules, and real-time communication technologies.

1.3 Objectives of the Project

The objectives of the project are as follows:

- To develop an IoT-Based Smart Doorbell System that enables real-time visitor detection and enhances home security by providing remote monitoring and verification through image capture and communication technologies.
- To implement automated door access control using embedded systems, along with safety features such as alerts and notifications, ensuring secure, efficient, and reliable home entry management.

1.4 Scope of the Project

The scope of the IoT-Based Smart Doorbell System focuses on developing a smart and automated home security solution that enables real-time visitor detection and remote access control. The system integrates both hardware and software components to ensure efficient monitoring and secure entry management.

The system includes functionalities such as visitor detection using an ultrasonic sensor and real-time image capture through a camera module. The captured images are sent to the homeowner via Telegram for remote verification and decision-making.

The system also provides automated door control, where access is granted based on user authorization. It ensures smooth door operation using a motor driver and maintains security through controlled access

The system enhances safety by incorporating alert mechanisms such as GSM notifications, buzzer, and LED indicators. It is suitable for applications in smart homes, apartments, and security-sensitive environments requiring reliable and intelligent entry management.

1.5 Project Introduction

Home security and controlled access are essential aspects of modern residential systems. However, traditional doorbell systems often suffer from limited functionality, lack of real-time monitoring, and absence of remote access, which can lead to security risks and inconvenience for homeowners.

The IoT-Based Smart Doorbell System aims to improve home security by integrating sensor-based detection, camera monitoring, and real-time communication technologies. In this system, an ultrasonic sensor is used to detect visitor presence, and a USB camera captures real-time images. These images are sent to the homeowner through Telegram, allowing remote verification and decision-making. Based on user authorization, the system enables automated door access and ensures secure entry management.

Important system operations such as visitor detection, image transmission, and access control are handled through embedded Python and Arduino-based hardware integration, ensuring efficient and reliable performance. Additional safety features such as GSM alerts, buzzer, and LED indicators enhance security by providing instant notifications during unauthorized access or emergency situations.

The system is developed using Arduino for hardware control and embedded Python for communication and automation. The frontend interaction is managed through Telegram integration, enabling real-time user communication. By combining IoT technology with automation, the system enhances security, improves response time, and provides a smart, reliable, and user-friendly solution for modern home entry management.

CHAPTER 2

LITERATURE SURVEY

In recent years, IoT-based smart home security systems have gained significant attention due to their ability to provide real-time monitoring, remote accessibility, and enhanced safety. S. R. Anupriya and V. Muthumanikandan [1] conducted a survey on IoT-based home security systems and emphasized their effectiveness in continuous monitoring and alert mechanisms. However, their study primarily focuses on general surveillance and lacks intelligent access control and automated door management features.

Several studies have explored smart doorbell systems integrated with image processing for visitor identification. A. B. Thabet and N. B. Amor [8] proposed a smart doorbell system using face recognition to identify authorized users. Although this approach enhances security, it involves complex image processing techniques and requires higher computational resources, making it less suitable for low-cost and embedded implementations. Similarly, A. Ennis et al. [10] developed a doorstep security system that captures visitor images and sends them for user verification. While this improves monitoring capabilities, it lacks automation in access control and real-time decision-making.

IoT-based home automation systems integrating multiple sensors have also been widely studied. M. Sayeduzzaman et al. [4] proposed an IoT-integrated home automation system with smart security features, offering real-time alerts and remote monitoring. However, their system does not include efficient visitor verification or automated door access mechanisms. Likewise, V. Rajeswari et al. [3] introduced a smart home security alert system for continuous supervision, but it mainly focuses on detection rather than controlled access and decision-making.

Communication-based notification systems have also been developed to enhance responsiveness. W. H. Park and Y. G. Cheong [7] proposed an IoT-based smart bell notification system that sends alerts to users through connected device. While this improve

Communication-based notification systems have also been developed to enhance responsiveness. W. H. Park and Y. G. Cheong [7] proposed an IoT-based smart bell notification system that sends alerts to users through connected devices

Advanced techniques such as machine learning and deep learning have also been applied to improve security systems. A. Rahim et al. [2] proposed a system combining anomaly detection and face recognition using logit-boosted CNN models to enhance security accuracy. Although the system achieves better detection performance, it is computationally intensive and not ideal for real-time embedded systems. Additionally, A. Baobaid et al. [9] explored hardware accelerators for real-time face recognition, highlighting the need for optimized processing; however, such solutions increase system cost and complexity.

Further studies have focused on improving accessibility and smart locking mechanisms. Y. K. Verma et al. [6] designed a smart doorbell system that enables remote monitoring and visitor notification. While their system improves user interaction, it lacks automated door control and integrated safety mechanisms such as multi-level alerts and emergency handling. Similarly, S. R. Pinjala and S. Gupta [11] proposed a remotely accessible smart lock system, but it does not provide comprehensive integration with visitor detection and communication systems.

Other IoT applications have also contributed to smart environments. M. N. Mostakim et al. [13] developed an intelligent home system with automated environmental control, focusing more on comfort than security. Similarly, L. Eras et al. [15] explored Bluetooth mesh-based smart building applications, emphasizing communication efficiency but lacking integrated security functionalities.

Overall, existing literature addresses various aspects such as visitor detection, remote monitoring, alert systems, and intelligent recognition. However, most systems either focus only on monitoring without access control, rely on complex and costly technologies, or lack proper integration of automation and communication The proposed system addresses these limitations by providing a cost-effective IoT-based smart doorbell

olution that integrates real-time visitor detection, image-based verification via Telegram, remote access control, and automated door operation. Additionally, it ensures better synchronization between hardware and software components, improves response time, and enhances overall system reliability, making it suitable for practical smart home applications

CHAPTER 3

DESIGN

The design phase of the IoT-Based Smart Doorbell System focuses on defining the structure, workflow, and functional components of the system. This chapter presents the overall system design, including the analysis of the existing system, proposed solution, and detailed working of the application. It also describes how different components such as sensors, camera module, embedded system, communication platform, and alert mechanisms work together to provide a secure and efficient home entry management system. The design ensures that the system meets all functional requirements while maintaining security, reliability, and real-time responsiveness.

3.1 Existing System

Traditional doorbell systems are simple and lack advanced security features. These systems rely on manual interaction, where the homeowner must be physically present to respond to visitors. There is no provision for remote monitoring, visitor verification, or automated access control, making them less effective in modern security scenarios.

These systems are also limited in functionality as they do not provide real-time alerts, image capture, or communication with the homeowner when they are away. Additionally, they cannot detect unauthorized access or suspicious activities effectively. Since there is no integration of intelligent technologies, the system fails to ensure safety, convenience, and efficient monitoring, leading to security risks and delayed responses.

3.1.1 Disadvantages of Existing System

1. Lack of real-time monitoring and remote access
2. No visitor verification or image capture mechanism
3. High dependency on manual interaction
4. Absence of automated door control and alert systems

3.2 Proposed System

The proposed system introduces an IoT-Based Smart Doorbell System that enhances home security, automation, and real-time monitoring. The system follows an integrated architecture where hardware components such as sensors, camera modules, and actuators are combined with embedded Python and Arduino-based control to provide efficient and secure entry management.

The system consists of key functionalities such as visitor detection, image capture, remote verification, and automated door control. When a visitor is detected using the ultrasonic sensor, the camera captures an image and sends it to the homeowner via Telegram. Based on user authorization, the system allows or denies access. The system also includes alert mechanisms to notify users during unauthorized access or emergency situations.

IoT technology is used to enable real-time communication between devices and the user, ensuring quick response and remote accessibility. The integration of sensors, communication modules, and automation ensures a reliable, secure, and efficient smart home solution.

3.2.1 Advantages of the Proposed System

1. Enhanced home security through real-time visitor detection and monitoring
2. Remote access and control using Telegram communication
3. Automated door operation reduces manual effort
4. Instant alerts for unauthorized access and emergency situations
5. Cost-effective and efficient solution using IoT and embedded systems

3.1 Project Flow

The project flow of the IoT-Based Smart Doorbell System represents the interaction between the visitor, homeowner, and the system. Initially, the ultrasonic sensor detects the presence of a visitor near the door. Once detected, the camera captures an image and sends it to the homeowner through Telegram for verification.

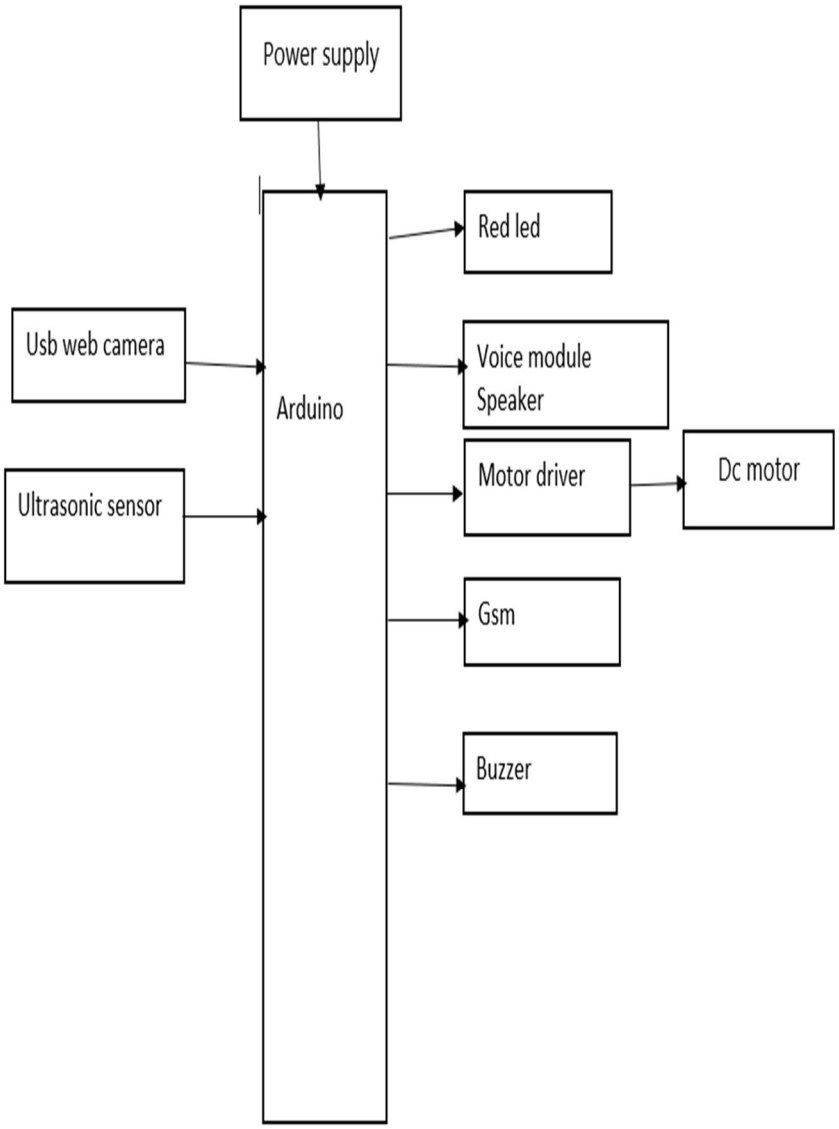

Fig No 3.1: Block diagram of proposed system

Initially, the process begins when the system is powered on and continuously monitors the entrance area using the ultrasonic sensor. When a visitor approaches the door, the sensor detects their presence within a predefined range. Once detection occurs, the system activates the camera module to capture an image of the visitor. The captured image is then sent to the homeowner through Telegram for real-time verification.

The homeowner reviews the received image and decides whether to allow or deny access. If access is denied or no response is received, the system remains locked and may trigger alert mechanisms such as a buzzer or LED indicator. If the homeowner grants access, the system activates the motor driver to unlock and open the door. The door remains open for a fixed duration and then automatically closes to maintain security.

All system activities such as visitor detection, image transmission, and access control are processed in real time, ensuring efficient and secure entry management. The integration of communication and automation ensures improved response time, reliability, and user convenience.

3.2 Functional Requirements

The IoT-Based Smart Doorbell System is designed to manage visitor detection and access control efficiently using integrated hardware and software components. The system includes key functionalities such as detection, communication, verification, and automation.

The system must detect visitors, capture images, send them via Telegram, and allow remote access control with alert mechanisms for security. It should also ensure real- time communication and quick response.

The system must continuously monitor the entrance using sensors and detect visitor presence accurately. It should capture images and send them to the homeowner in real time for quick verification. The communication between components must be fast and reliable to avoid delays.

The system must enable remote access control while ensuring reliable operation and proper error handling.

Examples of Functional Requirements:

- The system must detect visitors using an ultrasonic sensor

- The system must capture and send visitor images via Telegram
- The user must be able to allow or deny access remotely
- The system must automatically open and close the door upon authorization
- The system must provide alerts using buzzer, LED, or GSM module
- The system must maintain real-time communication and response

3.3 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system such as performance, security, and usability.

Examples of Non-Functional Requirements:

1. The system must ensure reliable and continuous operation of sensors and hardware components
2. The system must provide secure communication between devices and the user
3. The system should respond to visitor detection and user commands in real time
4. The system must provide a user-friendly interaction through Telegram
5. The system should be available with minimal downtime
6. The system must ensure safety and prevent unauthorized access
7. The system should be scalable for different home environments
8. The system must maintain proper error handling and system stability

3.4 Hardware Requirements

Sensor Module	: HC-SR04 Ultrasonic Sensor
Microcontroller	: Arduino Uno
Camera	: USB Camera Module
Communication	: GSM/WI-FI Module
Alert Module	: Buzzer & LED
Motor Driver	: Relay Module

3.5 Software Requirements

Operating System	: Windows 7/8/10 or above

Programming Languages	: Python
Software Tools	: Arduino IDE
Simulation Tool	: Proteus
Libraries	: Ultrasonic Sensor Library

3.6 System Architecture

The architecture of the IoT-Based Smart Doorbell System follows a layered approach to ensure efficient processing, real-time communication, and secure home access control. The system consists of four main layers: presentation layer, application layer, hardware layer, and communication layer. These layers work together to provide a complete and reliable smart home security solution.

The presentation layer serves as the user interface through which the homeowner interacts with the system using a communication platform such as Telegram. This layer allows the user to receive real-time notifications, view captured images of visitors, and send commands to allow or deny access. It acts as the point of interaction between the user and the system, ensuring ease of use and accessibility from remote locations.

The application layer acts as the core processing unit of the system and is implemented using embedded Python along with Arduino programming. It handles all system logic, including visitor detection processing, image handling, command execution, and control of hardware components. This layer processes input from sensors, sends data to the user, receives user responses, and controls the door operation accordingly. It ensures smooth coordination between all components and manages overall system functionality.

The hardware layer is responsible for the physical components of the system. It includes the ultrasonic sensor for detecting visitor presence, the camera module for capturing images, the Arduino microcontroller for processing, and the motor driver for controlling door movement. Additional components such as buzzer, LED, and GSM module provide alert mechanisms for unauthorized access or emergency situations. This layer ensures accurate sensing, actuation, and real-time system response.

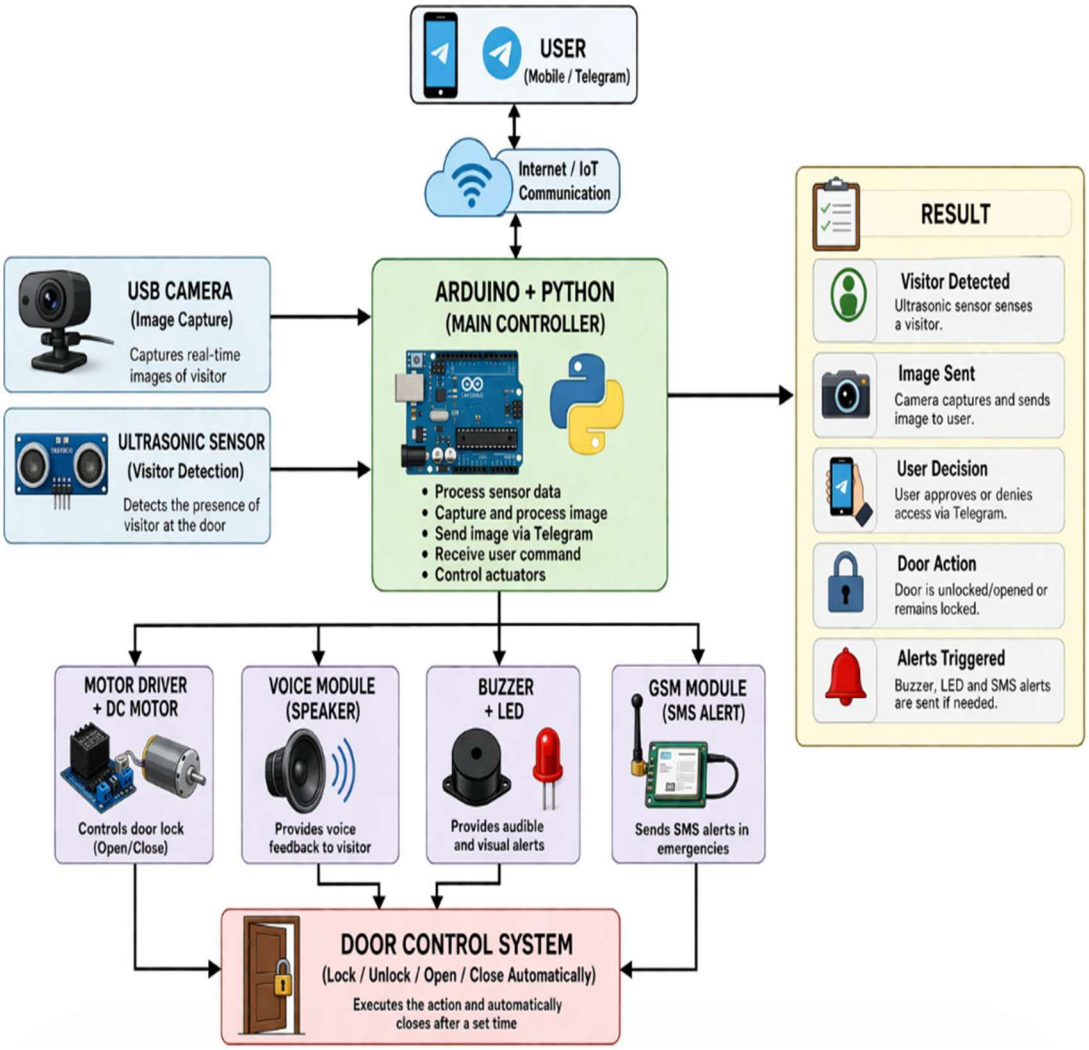

Fig No 3.2: System Architecture

The communication layer enables data exchange between the system and the user. It uses internet-based communication through Telegram and optionally GSM for sending alerts. This layer ensures that notifications are delivered instantly and that user commands are received without delay. It plays a critical role in enabling remote monitoring and control of the system.

Overall, the system processes visitor detection through the hardware layer, handles logic in the application layer, communicates with the user through the communication layer, and provides interaction through the presentation layer. This

architecture ensures reliability, real-time performance, and enhanced security, making the system suitable for modern smart home applications.

3.7 Input Design

Input design is an important phase in system development that focuses on how data is entered into the system. It ensures that the input provided is accurate, valid, and easy to process.

In the IoT-Based Smart Doorbell System, input is mainly based on sensor data and user commands. The ultrasonic sensor continuously provides distance data to detect visitor presence. Once a visitor is detected, the system captures an image using the camera module. The homeowner provides input in the form of commands (allow or deny access) through Telegram.

Proper validation techniques are applied to ensure reliable system operation, such as accurate distance detection, proper image capture, and correct interpretation of user commands. This helps reduce errors and ensures smooth system functionality.

3.7.1 Objectives of Input Design:

- Ensure accurate detection of visitors using sensor input
- Provide simple and user-friendly command input through Telegram
- Reduce errors through proper validation of sensor and user inputs
- Maintain consistency in data processing
- Improve system reliability and efficiency

3.8 Output Design

Output design defines how the system presents processed information to the user in a clear and understandable manner.

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In this system, outputs are provided through Telegram notifications and hardware indicators. The homeowner receives real-time images of visitors along with options to allow or deny access. The system also provides outputs such as door status (opened/closed), alert messages, and notifications during unauthorized access or emergency situations.

Additionally, visual and audio outputs such as LED indicators and buzzer alerts are used to provide immediate feedback. These outputs ensure that users are well- informed about system activities and can respond effectively.

3.8.1 Objectives of Output Design:

- Provide clear and real-time notifications to the user\
- Provide clear and real-time notifications to the user
- Ensure easy understanding of system alerts and messages
- Provide effective audio and visual alerts
- Maintain a user-friendly interaction system

3.9 UML Diagrams

Unified Modeling Language (UML) is a standardized modeling language used to visualize, design, and document the structure and behavior of a system. It provides a clear representation of system components, their relationships, and data flow, making complex systems easier to understand and analyze.

UML plays an important role in system development by helping designers and developers define system architecture, interactions, and workflows. It supports both structural and behavioral modeling, ensuring that all system functionalities are properly represented. UML diagrams also help in identifying potential issues early and improving communication between developers.

In the IoT-Based Smart Doorbell System, UML diagrams are used to represent system design, user interactions, and operational flow. These diagrams illustrate how components such as sensors, camera, Arduino, and communication modules interact with the user.

Overall, UML diagrams simplify system understanding and enhance documentation quality.

3.9.1 Use Case Diagram

The use case diagram represents the interactions between the user (homeowner) and the system in the IoT-Based Smart Doorbell System. It illustrates the functionalities available to the user and how they interact with the system to perform various operations, as shown in Figure 3.3.

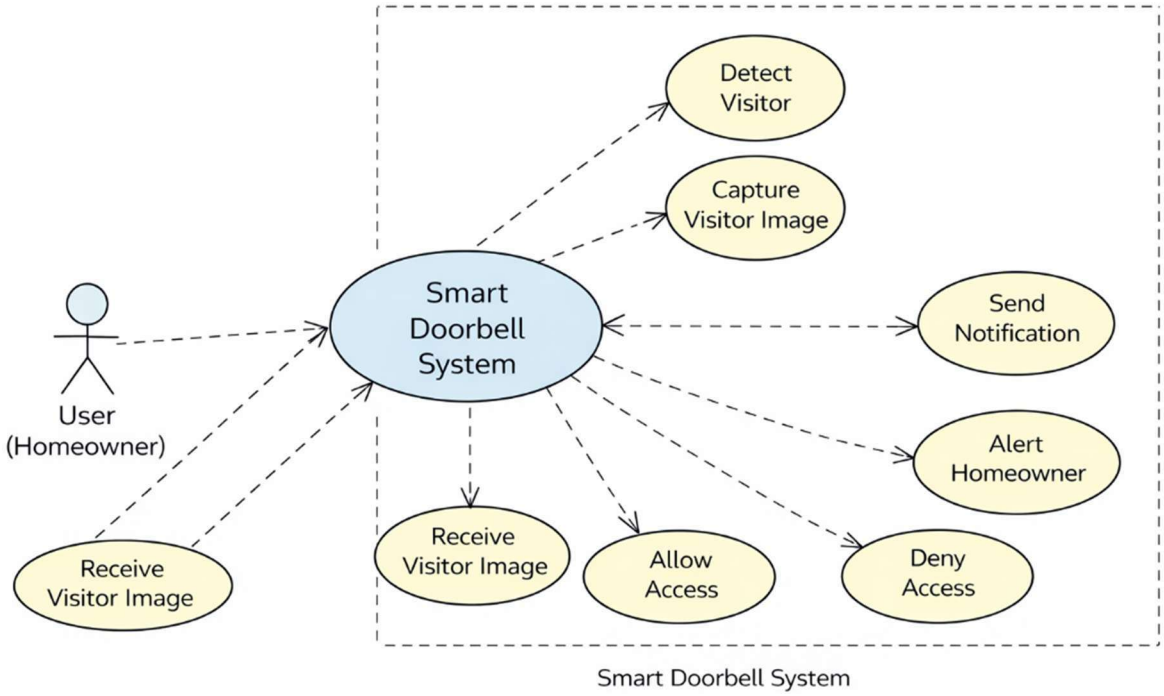

Fig No 3.3: Use case diagram

In this system, the user interacts with the system through a communication platform such as Telegram. When a visitor approaches the door, the system automatically detects their presence using the ultrasonic sensor and captures an image using the camera module. The image is then sent to the user for verification. After receiving the notification, the user can view the visitor image and decide whether to allow or deny access. If access is granted, the system unlocks and opens the door automatically, and closes it after a fixed duration. The user can also receive alert notifications in case of unauthorized access or emergency situations.

The system performs several automated actions such as visitor detection, image capture, notification sending, and door control without manual intervention. It also activates alert mechanisms like buzzer, LED, and GSM notifications when required. The interaction between the user and the system is mainly based on real-time communication and decision-making.

Overall, the use case diagram clearly shows the roles and interactions between the user and the system. It highlights how the system supports visitor detection, remote verification, access control, and alert handling, ensuring a secure and efficient smart home entry management process.

3.9.2 Class Diagram

The class diagram represents the structural design of the IoT-Based Smart Doorbell System by illustrating the main classes, their attributes, methods, and relationships, as shown in Figure 3.4. It provides a clear understanding of how different components of the system are organized and interact with each other.

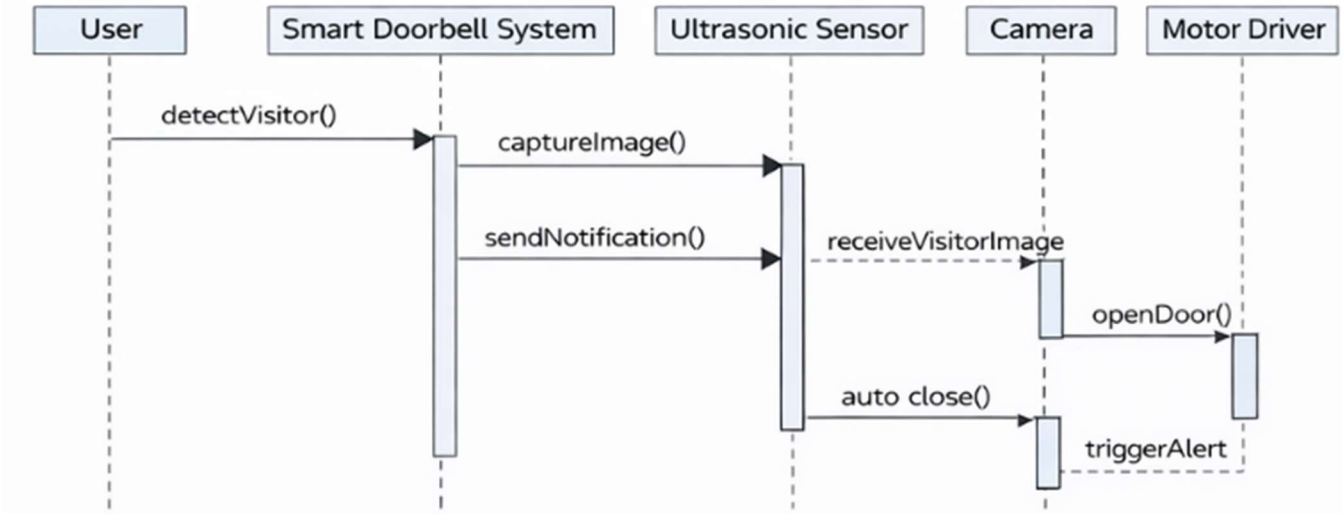

Fig No 3.4: Class diagram

The **User class** contains attributes such as user_id, name, and contact details, which represent the homeowner interacting with the system. This class is responsible for receiving notifications and sending commands to control access.

The **Visitor class** includes attributes such as visitor_id, image, and detection time. It is responsible for storing information about detected visitors and captured images.

The **DoorControl class** includes attributes such as door_status and methods like openDoor() and closeDoor(). It controls the physical operation of the door based on user input.

The relationships between these classes show that a user interacts with multiple visitor records, and each visitor detection triggers door control actions. This structure ensures efficient system operation and proper coordination between components.

3.9.3 Sequence Diagram

The sequence diagram illustrates the interaction between the user, system, and hardware components in a time-ordered manner, showing how operations are executed step by step in the IoT-Based Smart Doorbell System, as shown in Figure 3.5.

The process begins when the ultrasonic sensor detects the presence of a visitor near the door. The system then activates the camera to capture the visitor image. The captured image is sent to the homeowner through Telegram for verification.

The homeowner reviews the image and sends a command to either allow or deny access. The system processes the response and performs the corresponding action. If access is granted, the motor driver opens the door and automatically closes it after a fixed duration. If access is denied, the system triggers alert mechanisms such as buzzer and LED.

The sequence diagram clearly demonstrates the flow of interactions between the user and system components, ensuring real-time communication and efficient access control.

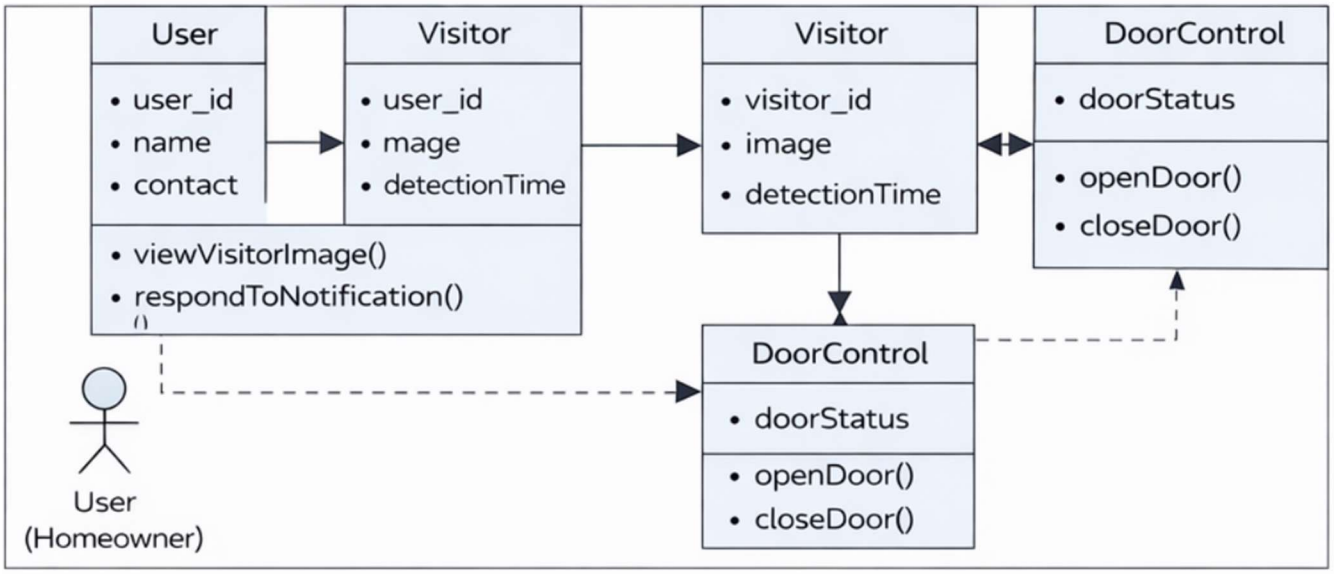

Fig No 3.5: Sequence diagram

3.9.4 Activity Diagram

The activity diagram represents the workflow of the IoT-Based Smart Doorbell System by illustrating the sequence of activities performed by the system and the user, as shown in Figure 3.6. It provides a clear view of how the system operates from start to end.

The process starts with visitor detection using the ultrasonic sensor. Once detected, the system captures the image and sends it to the user. The user then verifies the visitor and makes a decision.

Based on the decision, the system either allows access by opening the door or denies access and triggers alerts. The door automatically closes after a specific time if opened. The diagram shows a continuous loop of monitoring and response.

The activity diagram highlights the complete workflow, ensuring smooth coordination between detection, verification, and action.

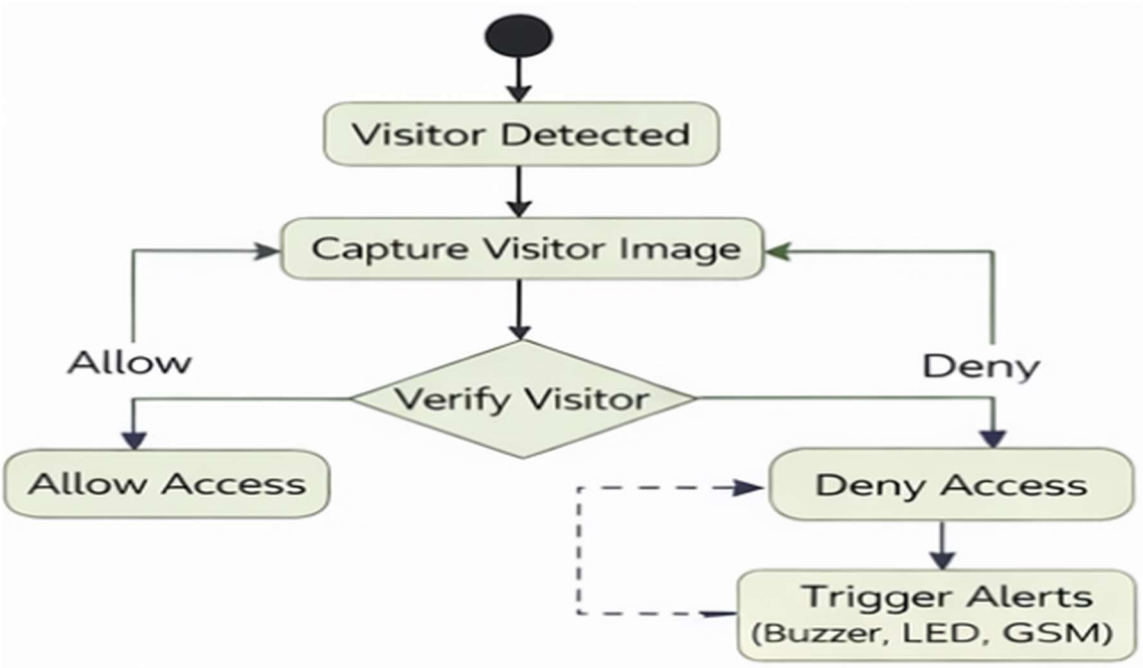

Fig No 3.6: Activity diagram

3.9.5 Component Diagram

The component diagram represents the overall structure of the IoT-Based Smart Doorbell System by illustrating the major system components and their interactions, as shown in Figure 3.7. It provides a high-level view of system organization.

The system consists of components such as the User Interface (Telegram), Arduino Controller, Sensor Module, Camera Module, and Alert System.

The user interface communicates with the system through internet-based communication. The Arduino acts as the central controller, processing inputs from the sensor and camera. The alert system provides notifications using buzzer, LED, and GSM module.

The interaction between these components ensures efficient data flow and system functionality, providing a reliable smart doorbell solution.

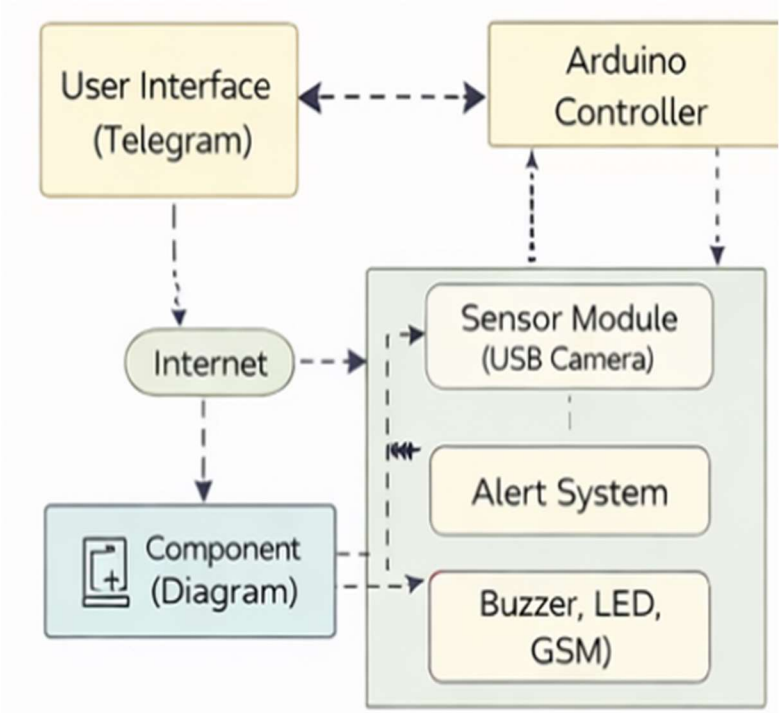

Fig No 3.7: Component diagram

3.10 Data Flow Diagram (DFD)

The Data Flow Diagram represents how data moves within the IoT-Based Smart Doorbell System, showing the interaction between the user, system processes, and hardware components, as illustrated in Figure 3.8. It provides a clear understanding of data processing and communication.

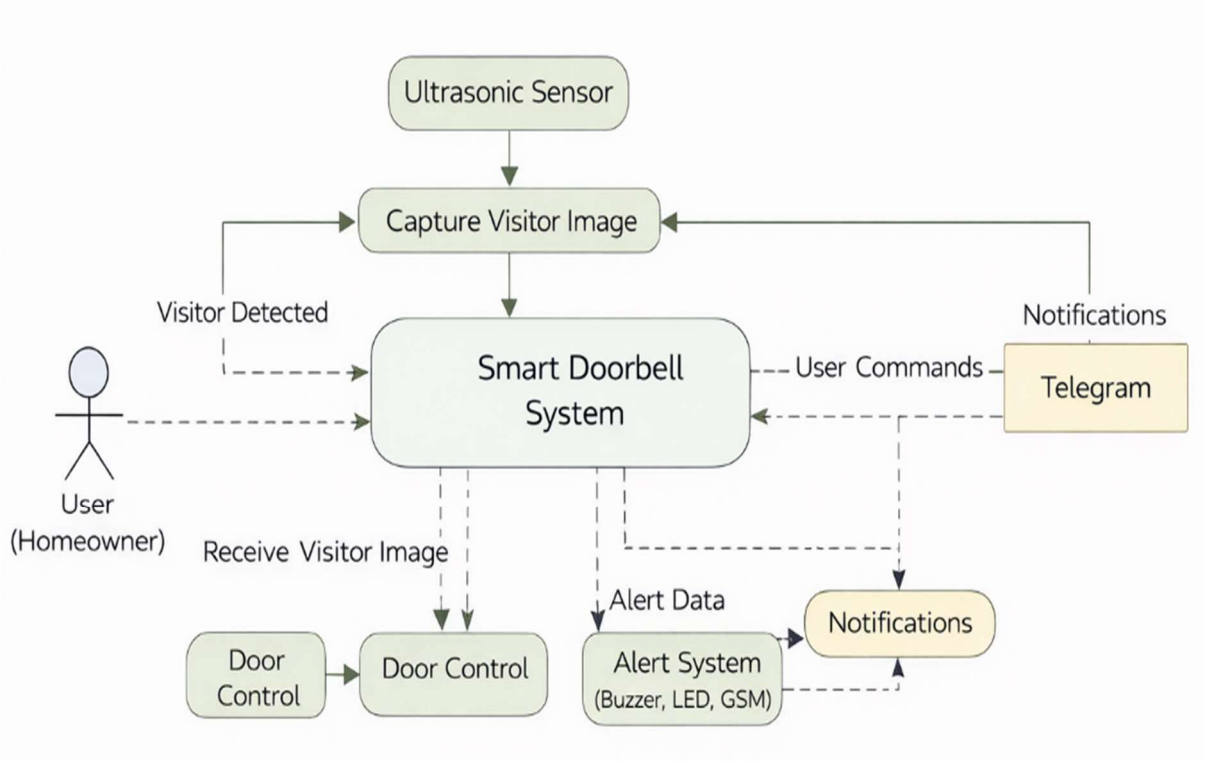

Fig No 3.8: Data flow diagram

The system processes data such as sensor input, captured images, and user commands. The ultrasonic sensor provides input data, which is processed by the system to detect visitors. The camera captures images, which are sent to the user via Telegram.

User responses are received and processed to control door operations. Alert data is generated and sent through buzzer, LED, or GSM module when necessary.

Overall, the Data Flow Diagram illustrates how data is captured processed, transmitted, and used for decision-making, ensuring smooth and efficient system operation

CHAPTER 4

IMPLEMENTATION

The implementation phase focuses on transforming the system design into a working application by integrating various hardware and software components. This phase involves developing sensor-based detection, camera integration, communication modules, and automation mechanisms required for secure and efficient home entry management. Each module is implemented and tested to perform specific functionalities, contributing to the overall performance, reliability, and responsiveness of the system.

4.1 Methodology

The development of the IoT-Based Smart Doorbell System follows an integrated architecture that combines embedded systems with real-time communication technology to ensure efficient and secure operation.

The system is implemented using Arduino for hardware control and embedded Python for processing and communication. The ultrasonic sensor is used for continuous monitoring and detecting visitor presence, while the camera module captures real-time images. These images are processed and transmitted to the homeowner through Telegram, enabling remote monitoring and verification.

During operation, the system also includes alert mechanisms such as buzzer, LED, and GSM module to notify users in case of unauthorized access or emergency situations. All system activities such as detection, communication, and door control are handled in real time, ensuring quick response and efficient functioning.

The development process includes proper synchronization between hardware and software, along with validation, error handling, and testing to ensure system reliability and safety. By combining IoT technology with automation and communication, the methodology ensures improved security, faster response time, and efficient home entry management.

4.2 User Module

The User Module is designed to handle all activities performed by the homeowner in the system. It provides a simple and user-friendly interface that allows the user to interact with the smart doorbell system efficiently and securely.

1. **User Interaction and Access Control**
   - The user receives real-time notifications when a visitor is detected at the door
   - The user can view captured images of visitors and make decisions to allow or deny access remotely through Telegram.
2. **Visitor Monitoring**
   - The system enables the user to monitor visitor activity in real time.
   - Users can view images of visitors and stay informed about who is at their door even when they are not physically present.
3. **Door Control Functionality**
   - Users can view images of visitors and stay informed about who is at their door even when they are not physically present.
   - Based on the user's response, the system automatically opens or keeps the door locked.
4. **Alerts and Notifications**
   - The user receives instant alerts in case of unauthorized access or suspicious activity.
   - Notifications are sent through Telegram and optionally via GSM for emergency situations**.**
5. **System Feedback and Status**
   - The system provides feedback to the user regarding door status (opened/closed) and system actions.
   - The user is continuously updated about system operations to ensure transparency and control.

6. **Security and Convenience**
   - The module ensures secure communication between the system and the user.
   - It provides convenience by enabling remote monitoring and control, improving overall home security.
7. **Feedback Submission**
   - Users can submit feedback or complaints regarding the system.
   - This helps in improving system performance and user experience.
8. **Profile Management**
   - Users can manage basic settings such as notification preferences and contact details.
9. **Logout**
   - Users can securely disconnect from the system interface when required
   - This prevents unauthorized access and ensures system safety.

4.3 Admin Module

The Admin Module is responsible for monitoring, maintaining, and controlling the overall operation of the Smart Doorbell System. It ensures proper system functioning, validation, and security management.

1. **Admin Access**
   - Admin has control over system configuration and ensures all components function properly.
2. **System Monitoring**
   - Admin monitors visitor detection, alerts, and door operations in real time.
3. **Device Management**
   - Admin manages sensors, camera, and communication modules for proper connectivity.
4. **Payment Monitoring**
   - Admin ensures alert systems like buzzer, LED, and GSM work correctly during emergencies.
5. **System Maintenance**
   - Admin performs system checks, handles errors, and ensures smooth operation.

4.4 System Implementation

The system implementation plays a key role in ensuring security, automation, and real-time communication. Embedded programming using Arduino and Python is used to handle critical operations such as visitor detection, image capture, and door control.

When a visitor is detected, the system captures an image and sends it to the homeowner through Telegram. Based on the user's response, the system performs actions such as unlocking or keeping the door closed. This ensures secure and controlled access to the home.

The integration between hardware and communication modules is achieved using embedded Python and network connectivity, which facilitates real-time interaction between the system and the user. This implementation ensures reliable, secure, and efficient home entry management.

4.5 Data Handling Implementation

The integration between hardware and communication modules is achieved using embedded Python and network connectivity, which facilitates real-time interaction between the system and the user. This implementation ensures reliable, secure, and efficient home entry management.

Temporary data such as visitor images and system responses are processed in real time to ensure quick decision-making. The system ensures proper synchronization between components to maintain accuracy and reliability

By combining sensor-based detection with real-time communication, the system ensures efficient data handling, quick response, and improved overall system performance.

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CHAPTER 5

TESTING

Testing is a crucial phase in the development of the IoT-Based Smart Doorbell System, ensuring that the system functions correctly and meets all specified requirements. It helps in identifying errors, improving performance, and ensuring system reliability. Various testing techniques are applied to validate both functional and non- functional aspects of the system, ensuring that each component works properly and that the entire system operates smoothly under different conditions.

5.1 System Testing

System testing follows a structured methodology to verify individual components as well as the complete system. It ensures that all modules work correctly both independently and in integration with other components. The following testing techniques are used in the system:

5.1.1 Unit Testing

Unit testing involves testing individual components of the system separately to ensure that each module functions correctly. In this project, components such as ultrasonic sensor detection, camera image capture, Telegram communication, and door control are tested independently. This helps in identifying errors at an early stage and improves system reliability.

5.1.2 Integration Testing

Integration testing focuses on verifying the interaction between different components of the system. It ensures that the sensor, camera, Arduino, and communication modules work together seamlessly. This testing helps identify issues related to data flow, communication, and synchronization between hardware and software.

5.1.3 Functional Testing

Functional testing is performed to verify that all system functionalities meet the specified requirements. It includes testing features such as visitor detection, image transmission, remote access control, and alert mechanisms. This ensures that the system behaves as expected and provides correct outputs for given inputs.

5.1.4 White Box Testing

White box testing involves examining the internal logic and control flow of the system. It focuses on testing embedded code, sensor processing logic, and communication handling. This ensures that all internal operations are performed correctly and efficiently.

5.1.5 Black Box Testing

Black box testing is performed without knowledge of the internal code. The system is tested by providing inputs such as visitor presence and user commands, and verifying the outputs like alerts and door operations. This ensures the system meets user requirements and functions correctly.

5.2 Features to be Tested

The following key features of the system are tested to ensure proper functionality:

The visitor detection functionality is tested to ensure accurate sensing using the ultrasonic sensor. The image capture and transmission feature is verified to confirm that visitor images are sent correctly through Telegram. The remote access control feature is tested to ensure that users can allow or deny access effectively.

The door control mechanism is tested to verify automatic opening and closing operations. Alert systems such as buzzer, LED, and GSM notifications are tested to

ensure proper functioning during unauthorized access or emergencies. System reliability, real-time communication, and error handling are also tested to ensure smooth and secure operation.

5.3 Testing Tools and Environment

The testing of the IoT-Based Smart Doorbell System is carried out using a combination of hardware and software tools. Arduino IDE is used for programming and testing the microcontroller, while embedded Python is used for communication and processing.

Telegram is used as the communication platform to test real-time notifications and user interaction. Hardware components such as ultrasonic sensor, camera module, motor driver, buzzer, and GSM module are tested in a real-time environment to ensure proper functionality.

The development and testing are performed using standard computing systems with necessary software tools installed. This setup ensures that all components of the system function correctly, efficiently, and reliably

In addition to the above, real-time testing is performed to evaluate the system's response speed and reliability under different conditions. Network connectivity is tested to ensure smooth communication between the system and Telegram without delays or failures. Hardware components are also tested under continuous operation to verify stability and performance. This ensures that the system can handle real-world scenarios effectively while maintaining accuracy, responsiveness, and overall system reliability.

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5.4 Test Cases

The following test cases are used to verify the correctness and functionality of the system:

Table 8.1: Test Cases

S. No	Test Case	Input	Expected Output	Actual Output	Result
1	Visitor Detection	Person near sensor	Visitor detected	Detected successfully	Pass
2	Image Capture	Visitor detected	Image captured	Image captured	Pass
3	Image Transmission	Capture image	Sent via Telegram	Sent successfully	Pass
4	Access Control	User Allow access	Door opens	Door opened	Pass
5	Access Denied	User denies access	Door remains closed	Door closed	Pass
6	Door Auto Close	Door opened	Closes after time	Closed correctly	Pass
7	Alert System	Unauthorized access	Buzzer/LED activated	Alert triggered	Pass
8	Communication	Unauthorized access	System responds correctly	Response successful	Pass

CHAPTER 6

RESULTS

This chapter presents the outcomes of the IoT-Based Smart Doorbell System after implementation. It focuses on evaluating the system's performance, validating real-time communication, and analyzing improvements over traditional doorbell systems. The results demonstrate the effectiveness, reliability, and enhanced security achieved through the integration of IoT technology.

6.1 Performance Analysis

The performance analysis evaluates how efficiently the system performs various operations such as visitor detection, image capture, notification delivery, and door control. It highlights the system's ability to provide real-time responses, automate processes, and improve overall home security compared to traditional systems.

6.1.1 Functional Analysis

The implemented system successfully performs all key functionalities in an automated manner. The ultrasonic sensor accurately detects the presence of a visitor, and the camera captures real-time images. These images are immediately sent to the homeowner through Telegram, allowing remote monitoring and quick decision-making.

The system enables the user to allow or deny access based on the received image. If access is granted, the door opens automatically and closes after a predefined time. In case of unauthorized access or no response, the system triggers alert mechanisms such as buzzer, LED, and GSM notifications. All operations are performed efficiently, ensuring smooth workflow and minimal manual intervention.

6.1.2 System Validation

The system components are validated to ensure correct functioning of all operations. Sensor detection accuracy, image transmission, and communication through Telegram are tested to confirm reliability. The door control mechanism is verified to ensure proper opening and closing operations.

The system ensures real-time response with minimal delay between detection and notification. Alerts are triggered correctly during emergency conditions, and communication between hardware and software components is stable. This validation confirms that the system operates correctly and meets all functional requirements.

6.1.3 Comparative Performance Analysis

The comparative performance analysis evaluates the efficiency of the proposed IoT-Based Smart Doorbell System in comparison with traditional doorbell systems. The comparison is based on parameters such as response time, security level, and automation.

Comparison of Response Time

The response time refers to the duration required for the system to detect a visitor and notify the homeowner. As shown in Figure 6.1, the proposed IoT-Based Smart Doorbell System significantly reduces response time by automating key processes such as visitor detection, image capture, and instant notification through Telegram.

In the traditional system, the homeowner must be physically present to respond to the doorbell, which may lead to delays or missed visitors. There is no real-time notification or remote interaction, increasing response time and reducing efficiency.

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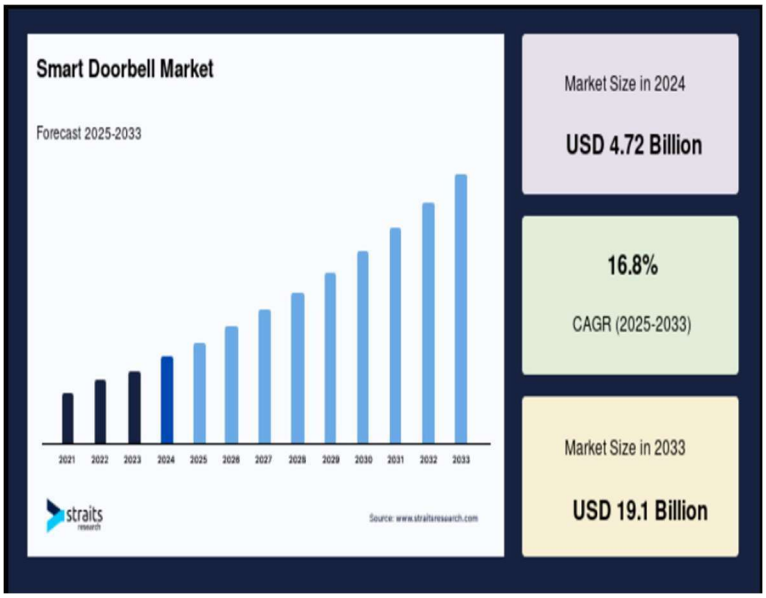

Fig No 6.1: Response Time Comparison

In contrast, the proposed system streamlines these operations by automatically detecting visitors and sending instant notifications to the user. This enables quick decision-making and immediate action, reducing delays and improving overall system efficiency.

Comparison of Notification Processing Time

Notification processing time refers to the time taken to detect a visitor, process the data, and send a notification to the homeowner.

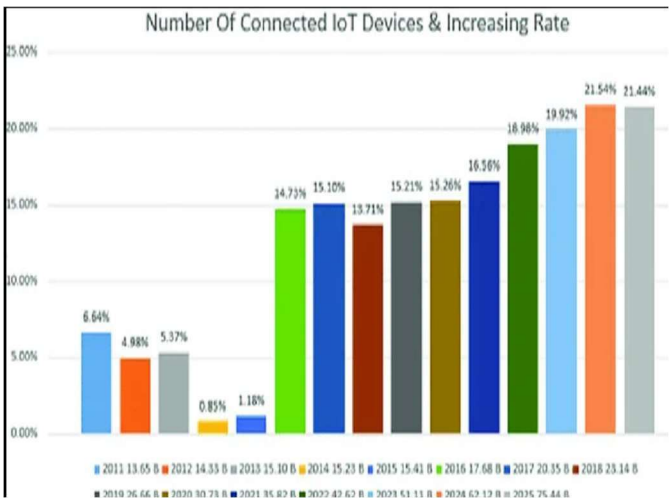

Fig No 6.2: Notification Processing Time Comparison

As illustrated in Figure 6.2, the traditional system experiences delays because it depends on manual response and physical presence of the homeowner. There is no automated mechanism to notify the user remotely, which increases the overall response delay.

In the proposed system, visitor detection, image capture, and notification sending are performed automatically using IoT components and Telegram integration. These operations are executed in real time without manual intervention, ensuring faster and consistent performance. As a result, the system maintains low notification processing time even with continuous usage, improving efficiency and user convenience.

Throughput Analysis

Throughput refers to the number of operations processed by the system within a given period. As shown in Figure 6.3, the proposed IoT-Based Smart Doorbell System demonstrates improved throughput compared to the traditional system.

Fig No 6.3: Throughput Analysis Comparison

This improvement is achieved because the system automates key operations such as visitor detection, image capture, notification delivery, and door control. In the traditional system, operations depend on manual interaction, which limits the number of actions handled within a given time.

6.2 OUTPUT SCREENS

This section presents the execution flow of the IoT-Based Smart Doorbell System through real-time output screens and hardware implementation. The images demonstrate how the system operates, including hardware setup, circuit design, detection process, and alert mechanism. These outputs confirm that the system is functioning correctly and all modules are integrated successfully.

Hardware Implementation

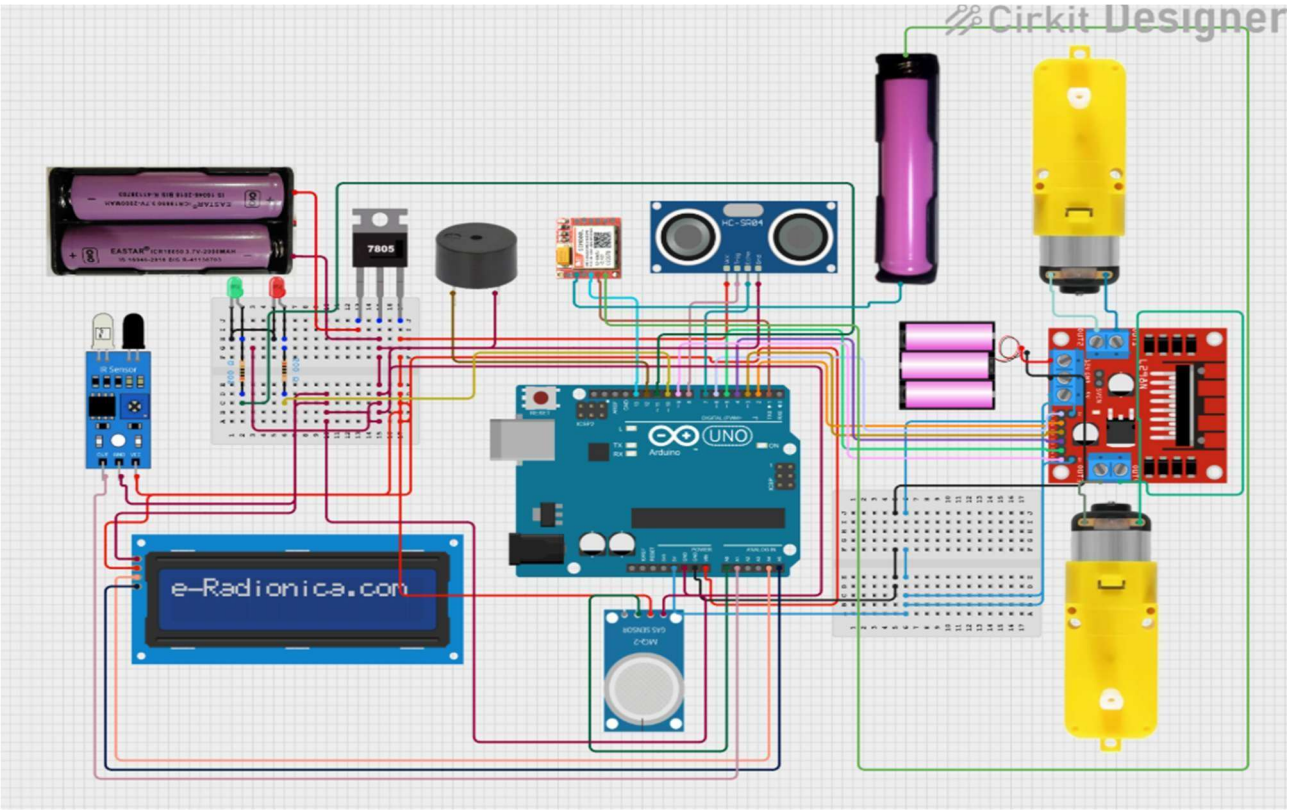

Fig No 6.4: Hardware Setup

The above figure shows the complete hardware implementation of the smart doorbell system. It includes Arduino Uno, GSM module, motor driver, buzzer, and other components assembled together. The setup ensures proper connectivity and real-time functioning of all modules.

Detection Screen:

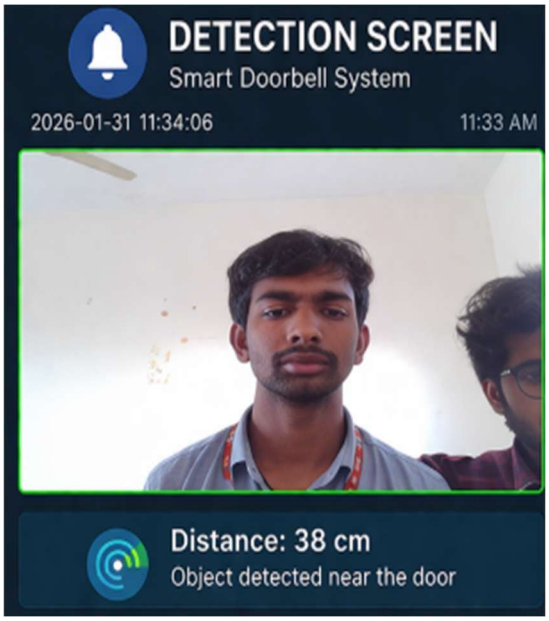

Fig No 6.5: Detection Screen

- Displays real-time distance values measured by the ultrasonic sensor
- Shows detection status (Visitor Detected / No Visitor)
- Provides continuous updates of sensor readings

Processing Screen:

Fig No 6.6: Processing Screen

- Displays the processing of sensor data in real-time
- Highlights when the distance is below the threshold
- Shows decision-making status of the system
- Indicates system activity and working condition

Alert Screen:

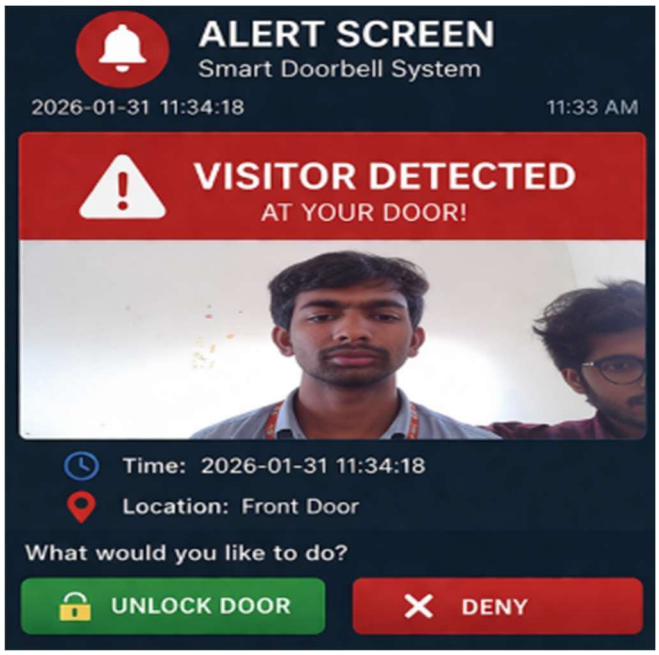

Fig No 6.7 Alert Screen

- Displays alert message: *"Visitor Detected at the Door"*
- Activates buzzer or doorbell sound
- Indicates successful detection and alert triggering
- Provides clear notification to the user

Result Screen:

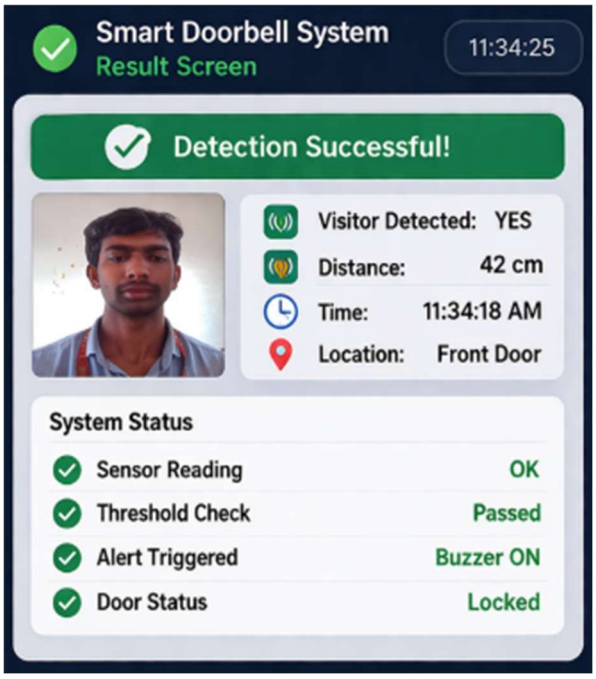

Fig No 6.8: Result Screen

- The system successfully detects visitors using the ultrasonic sensor
- Distance measurement is accurate and reliable
- The buzzer alert is triggered correctly when a visitor is detected
- Continuous monitoring ensures real-time operation
- The system reduces manual effort and improves home security

CONCLUSION

The IoT-Based Smart Doorbell System significantly improves home security and access management by integrating sensor-based detection, real-time communication, and automation. Unlike traditional doorbell systems, this system enables remote monitoring and control, ensuring that homeowners can verify visitors and make decisions from anywhere. The use of IoT technology ensures faster response, increased convenience, and enhanced safety.

The system effectively detects visitors using an ultrasonic sensor and captures real-time images through a camera module. These images are instantly transmitted to the user via Telegram, allowing quick verification. Based on user input, the system performs automated door control operations, reducing manual effort and improving efficiency. Additional alert mechanisms such as buzzer, LED, and GSM notifications further enhance security by providing instant warnings during unauthorized access or emergency situations.

The integration of hardware and software components ensures reliable system performance and smooth operation. The system minimizes delays, reduces human dependency, and provides a user-friendly interface for interaction. It demonstrates how IoT technology can be effectively applied to develop intelligent and automated home security solutions.

In a rapidly evolving smart home environment, this system provides a cost-effective, scalable, and practical solution for modern security needs. It not only addresses the limitations of traditional systems but also sets a foundation for future advancements in IoT-based automation. Overall, the project successfully achieves its objective of providing a secure, efficient, and user-friendly smart doorbell system for real-world applications.

FUTURE ENHANCEMENT

The proposed IoT-Based Smart Doorbell System can be further enhanced by incorporating several advanced features to improve functionality, scalability, and user experience.

In the future, the system can be extended to include facial recognition technology, allowing automatic identification of known visitors. This will enable the system to grant access to authorized individuals without manual intervention, improving convenience and security.

The system can also be enhanced by integrating a mobile application, which will provide a more user-friendly interface for monitoring, notifications, and door control. This will allow users to manage the system easily from their smartphones.

Advanced cloud integration can be implemented to store visitor images and logs securely. This will help in maintaining records for future reference and improve data accessibility from anywhere.

Security can be further improved by implementing advanced authentication methods such as two-factor authentication and encrypted communication to prevent unauthorized access.

The system can also be upgraded with AI-based anomaly detection to identify suspicious behavior and trigger alerts automatically. Integration with other smart home devices can further enhance automation and create a complete smart home ecosystem.

Overall, these enhancements will make the system more intelligent, secure, scalable, and suitable for real-world smart home applications.

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Publications related to project

Title: *Indian Patent Application Submission & Publication Status*

We are pleased to inform that our Indian patent application has been successfully filed.

The application is expected to be published within a timeframe of minimum 7 days to maximum 45 days from the date of filing.

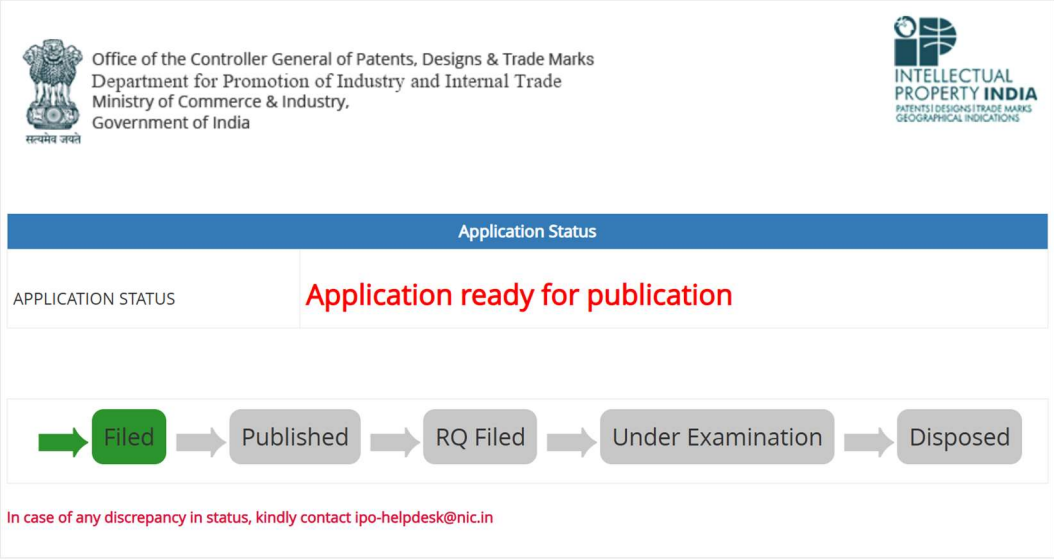
Application Details

Application Number: 202641037461

Publication Status: Pending (Will be published within 7–45 days)

Once the patent is published, kindly verify the application details using the official portal provided below:

Patent Application Status Link:
https://iprsearch.ipindia.gov.in/PublicSearch/PublicationSearch/ApplicationStatus



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Project CO-PO Mapping

Course Outcomes (COs)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1: Review the literature to understand the current state of art in the field of computer science	✓	✓	✓	✓	✓	-	-	-	-	✓	-	✓
CO2: Identify a problem and its scope for the proposed work in the selected field of study	✓	✓	✓	✓	-	-	-	-	-	-	-	✓
CO3: Develop appropriate mathematical model or prototype to solve the proposed problem	✓	✓	✓	✓	✓	-	-	-	-	-	-	-
CO4: Execute the proposed techniques with teamwork to solve the problem under consideration	✓	✓	✓	✓	✓	✓	-	-	✓	✓	✓	-
CO5: Evaluate the proposed techniques with adequate experimental analysis	✓	✓	✓	✓	✓	-	-	-	✓	✓	-	-
CO6: Prepare a dissertation following professional ethics	-	-	✓	-	-	✓	✓	✓	-	✓	✓	✓

Details of Project Team

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